

Linear Algebra

Saint Even, Slipper King

Source: <https://github.com/yunfeix2009/lin-alg-notes-ocw>

Contents

1	Preface	1
2	Vectors	2
2.1	Linear Combinations	2
2.2	Length and Dot Product	3
2.3	Cross Product	7
3	Matrices	10
3.1	Multiplication	10
3.2	Row Picture and Column Picture	11
3.2.1	out product	13
3.2.2	block	13
3.3	Transpose	14
3.4	Permutation Matrices	16
3.5	Difference and Sum Matrices	16
4	Linear Equations	19
4.1	Row Picture and Column Picture	19
4.2	Elimination	21
5	Vector Spaces and Subspaces	26
5.1	Spaces of Vectors	26
5.2	Null Spaces	28
5.3	Independence, Basis, Dimensions	29
5.4	Fundamental Subspaces	30
5.5	Matrix Spaces	31
6	Applications of Linear Algebra	33
6.1	Graphs, Networks, Incidence Matrices	33
A	Appendix	34

1 Preface

This is a set of notes to MIT OpenCourseWare 18.06, Linear Algebra, based on lectures by Gilbert Strang, recorded in Spring, 2005, along with other materials from <https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/> and

<https://web.mit.edu/18.06/www/>. Specifically, some problems from Gilbert Strang's book, Introduction to Linear Algebra, 4th edition is included. When the source of an example problem is not otherwise stated, it may be assumed that the problem is from the book.

Course description from OCW website: This is a basic subject on matrix theory and linear algebra. Emphasis is given to topics that will be useful in other disciplines, including systems of equations, vector spaces, determinants, eigenvalues, similarity, and positive definite matrices.

2 Vectors

Since the essence of linear algebra studies the operation linear transformation and the spaces they form, we first introduce the vectors, which linear combination is exerted upon. Then, we introduce to other useful operations on vectors.

2.1 Linear Combinations

Vectors are arguably the basic elements of study in linear algebra, rather than individual numbers, along with the space they form, known as the vector space. Therefore, beginning with an official introduction on vectors seems like a logical decision.

Similar to sets, vector is an ordered collection, often of real numbers, that could represent many things, including force, velocity, position, and any kind of data that describes the same object.

The two fundamental operations of vectors, like many other algebras, are addition and (scalar) multiplication.

When two vectors are of the same dimension, or the number of elements they hold, their sum is just the vector with each element the sum of the two vectors' corresponding elements. Additionally, when the addition defined for the elements is commutative, then vector addition is commutative.

Scalar multiplication is when a constant c is multiplied to a vector v , resulting in each element of cv being cv_i .

The combination of the two operations form one of the most important operations in linear algebra, *linear combination*, which is defined as $\sum cv$.

Observe that if v is non-zero, then cv forms a line. But not that obviously, the linear combination of two vectors not pointing in the same direction, independent, fill the entire plane.

Definition 2.1.1. Vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are independent iff no linear combinations of them, except $0\mathbf{u} + 0\mathbf{v} + 0\mathbf{w}$, results in a zero vector. Otherwise, they are said to be dependent.

Geometrically, when adding a vector to a set of independent vectors, then a new dimension is created in the space of all linear combinations, then the new set of vectors are independent, vice versa.

2.2 Length and Dot Product

Definition 2.2.1. For vectors $\mathbf{a} = (a_1, a_2, \dots, a_n)$ and $\mathbf{b} = (b_1, b_2, \dots, b_n)$, their **dot product** (sometimes referred to as the inner product) is defined by $\mathbf{a} \cdot \mathbf{b} := \sum_{i=1}^n a_i b_i$. The dot product is a scalar.

Definition 2.2.2. For a fixed vector with components real numbers $\mathbf{a} = (a_1, \dots, a_n)$, its norm, or length, is defined as, $\|\mathbf{a}\| = \sqrt{\sum_{i=1}^n a_i^2}$.

Notice that $\mathbf{a} \cdot \mathbf{a} = \|\mathbf{a}\|^2$

Also, it is easy to verify that this way of measuring distance is symmetric, positive definite, and satisfies the triangle inequality, meaning that this way of measuring distance is a metric on $\mathbb{R}^n$.

Theorem 2.2.3. The triangle inequality states that

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|.$$

Theorem 2.2.4. The Cauchy Schwarz Inequality states that

$$|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \cdot \|\mathbf{v}\|.$$

Equality case is achieved when $\mathbf{u} \parallel \mathbf{v}$.

Proof. This could be shown either by the fact that $|\cos \theta| \leq 1$ or with Lagrange Identity (see Theorem 2.3.5). $\square$

Definition 2.2.5. A unit vector $\hat{\mathbf{u}}$ is a unit vector iff $\|\hat{\mathbf{u}}\| = 1$, or $\hat{\mathbf{u}} \cdot \hat{\mathbf{u}} = 1$.

Easy to see that the unit vector in the same direction as $\mathbf{v}$ is $\hat{\mathbf{u}} = \frac{\mathbf{v}}{\|\mathbf{v}\|}$.

Theorem 2.2.6. For two given vectors of the same dimension, $\mathbf{a} = (a_1, \dots, a_n)$ and $\mathbf{b} = (b_1, \dots, b_n)$, $\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta$.

Proof. By the Law of Cosines, if $\mathbf{c} = \mathbf{a} - \mathbf{b}$, then

$$\|\mathbf{c}\|^2 = \|\mathbf{a}\|^2 + \|\mathbf{b}\|^2 - 2\|\mathbf{a}\| \|\mathbf{b}\| \cos \theta.$$

On the other hand,

$$\|\mathbf{c}\|^2 = \mathbf{c} \cdot \mathbf{c} = (\mathbf{a} - \mathbf{b}) \cdot (\mathbf{a} - \mathbf{b})$$

$$= \|\mathbf{a}\|^2 + \|\mathbf{b}\|^2 - 2(\mathbf{a} \cdot \mathbf{b}).$$

Comparing the two expressions yields

$$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta. \quad \square$$

Corollary 2.2.6.1. For two given vectors $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}$ is orthogonal to $\mathbf{b}$ iff $\mathbf{a} \cdot \mathbf{b} = 0$.

Example 2.2.7. (18.03 Pset 1 PartII p2) Let $\mathbf{u}, \mathbf{v}$ be vectors.

(a) Show that

$$\frac{1}{4}(\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2) = \mathbf{u} \cdot \mathbf{v}.$$

(b) Given two non-zero vectors $\mathbf{u}$ and $\mathbf{v}$, find the unit vector which bisects the smaller angle between them.

Solution. (a) We proceed by expanding the left side: LHS = $\frac{1}{4}((\mathbf{u}^2 + 2\mathbf{u} \cdot \mathbf{v} + \mathbf{v}^2) - (\mathbf{u}^2 - 2\mathbf{u} \cdot \mathbf{v} + \mathbf{v}^2)) = \mathbf{u} \cdot \mathbf{v}$.

(b) Define $\hat{\mathbf{x}} := \frac{\mathbf{x}}{\|\mathbf{x}\|}$. The answer is

$$\frac{\hat{\mathbf{u}} + \hat{\mathbf{v}}}{\|\hat{\mathbf{u}} + \hat{\mathbf{v}}\|^2}$$

□

Example 2.2.8. (Textbook Section 1.2 p28) Can three vectors in the xy -plane have

$$\begin{cases} \mathbf{u} \cdot \mathbf{v} < 0 \\ \mathbf{u} \cdot \mathbf{w} < 0 \\ \mathbf{v} \cdot \mathbf{w} < 0 \end{cases} \quad ?$$

Solution. Yes, for example, when the vectors are the vertices of an equilateral triangle centered at the origin. In fact, this is true whenever the angles between pairwise vectors are all greater than 90° .

However, we provide a false proof for no such existence. Multiplying the first two inequalities,

$$\begin{aligned} (\mathbf{u} \cdot \mathbf{v}) \cdot (\mathbf{u} \cdot \mathbf{w}) &= \mathbf{u} \cdot \mathbf{w} \cdot (\mathbf{v} \cdot \mathbf{v}) \\ &= \mathbf{u} \cdot \mathbf{w} \cdot \|\mathbf{v}\|^2 > 0 \\ \implies \mathbf{u} \cdot \mathbf{w} &> 0, \end{aligned}$$

contradiction. This proof wrongly assumes that dot product is associative, which fails as *dot product produces a scalar*. □

Remark. This idea is readily generalizable to $\mathbb{R}^n$, shown by the following theorem.

Theorem 2.2.9. *In $\mathbb{R}^n$, at most $n + 1$ vectors can have pairwise negative dot product.*

Proof. (generated by LLM's, at the time I was writing this section, I couldn't prove this).

We prove both the upper bound and that it is attainable.

For the upper bound, suppose $\mathbf{v}_1, \dots, \mathbf{v}_m \in \mathbb{R}^n$ satisfy

$$\mathbf{v}_i \cdot \mathbf{v}_j < 0$$

whenever $i \neq j$. We claim that $m \leq n + 1$.

Assume for contradiction that $m \geq n + 2$. Then these m vectors, viewed as points in $\mathbb{R}^n$, are affinely dependent. Therefore there exist scalars $a_1, \dots, a_m$, not all zero, such that

$$\sum_{i=1}^m a_i \mathbf{v}_i = \mathbf{0} \quad \text{and} \quad \sum_{i=1}^m a_i = 0.$$

Since the coefficients sum to 0 and are not all zero, some are positive and some are negative. Let

$$P = \{i : a_i > 0\} \quad \text{and} \quad N = \{j : a_j < 0\}.$$

Moving the negative terms to the other side gives

$$\sum_{i \in P} a_i \mathbf{v}_i = \sum_{j \in N} (-a_j) \mathbf{v}_j.$$

Call this common vector $\mathbf{x}$. Then

$$\mathbf{x} \cdot \mathbf{x} = \left(\sum_{i \in P} a_i \mathbf{v}_i \right) \cdot \left(\sum_{j \in N} (-a_j) \mathbf{v}_j \right) = \sum_{i \in P} \sum_{j \in N} a_i (-a_j) (\mathbf{v}_i \cdot \mathbf{v}_j).$$

For every $i \in P$ and $j \in N$, we have $a_i > 0$, $-a_j > 0$, and $\mathbf{v}_i \cdot \mathbf{v}_j < 0$, so every term in the double sum is negative. Hence

$$\mathbf{x} \cdot \mathbf{x} < 0,$$

which is impossible because $\mathbf{x} \cdot \mathbf{x} = \|\mathbf{x}\|^2 \geq 0$. This contradiction proves that $m \leq n + 1$.

To see that $n + 1$ is possible, take the vertices of a regular simplex centered at the origin. Concretely, in $\mathbb{R}^{n+1}$ let

$$\mathbf{u}_i = \mathbf{e}_i - \frac{1}{n+1}(1, 1, \dots, 1), \quad i = 1, \dots, n+1.$$

These vectors lie in the hyperplane

$$x_1 + \dots + x_{n+1} = 0,$$

which is an n -dimensional subspace, so we may regard them as vectors in $\mathbb{R}^n$. For $i \neq j$,

$$\mathbf{u}_i \cdot \mathbf{u}_j = -\frac{1}{n+1} < 0.$$

Thus $n+1$ vectors with pairwise negative dot products do exist, and so the maximum possible number is $n+1$. $\square$

Example 2.2.10. Pick any three numbers such that $x + y + z = 0$. Find the angle between vectors $\mathbf{v} = (x, y, z)$ and $\mathbf{w} = (z, x, y)$, θ .

Solution.

$$\begin{aligned} (x + y + z)^2 &= x^2 + y^2 + z^2 + 2(xz + xy + yz) \\ &= \frac{1}{2}(\|\mathbf{v}\|^2 + \|\mathbf{w}\|^2) + \mathbf{v} \cdot \mathbf{w} = 0. \end{aligned}$$

However,

$$\cos \theta = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\|\|\mathbf{w}\|} = -\frac{1}{2}.$$

Thus, $\theta = 120^\circ$. $\square$

Example 2.2.11. Find four pairwise perpendicular unit vectors with all components equal to $\frac{1}{2}$ or $-\frac{1}{2}$.

Solution. From either $\mathbb{R}^3$ includes at most 3 pairwise perpendicular directions or that the components have absolute value $\frac{1}{2}$, we are searching in $\mathbb{R}^4$. Each vector must be $\frac{1}{2}(\pm 1, \pm 1, \pm 1, \pm 1)$. Assume $\frac{1}{2}(1, 1, 1, 1)$ is in the solution, then the rest three could be found, resulting in

$$\begin{cases} \frac{1}{2}(1, 1, 1, 1) \\ \frac{1}{2}(1, -1, 1, -1) \\ \frac{1}{2}(1, 1, -1, -1) \\ \frac{1}{2}(1, -1, -1, 1) \end{cases}.$$

$\square$

2.3 Cross Product

With Definition 2.2.1, we have the means to often conveniently express angles and lengths in terms of vectors. However, there is another important concept that we are unable to grapple with using dot product. And that is area. For which we seek help from the use of (2.3.4).

Theorem 2.3.1. *The area of the triangle formed by two vectors with the same dimensions, $\mathbf{a}$ and $\mathbf{b}$ is $\det \begin{pmatrix} a_1 & a_2 \\ b_1 & b_2 \end{pmatrix}$*

Proof. Let θ denote the angle between $\mathbf{a}$ and $\mathbf{b}$. Then geometrically, we have the area to be

$$\frac{1}{2} \|\mathbf{a}\| \|\mathbf{b}\| \sin \theta.$$

Therefore, we aim to find $\sin \theta$ in terms of $\mathbf{a}$ and $\mathbf{b}$.

Let $\mathbf{a}'$ be the image of rotation of $\mathbf{a}$ for $\frac{\pi}{2}$ in the direction towards $\mathbf{b}$. Let θ' denote the angle between $\mathbf{a}'$ and $\mathbf{b}$.

We have

$$\|\mathbf{a}\| \|\mathbf{b}\| \sin \theta = \|\mathbf{a}'\| \|\mathbf{b}\| \cos(\theta') = \mathbf{a}' \cdot \mathbf{b}.$$

Let $\mathbf{a}$ have components (a_1, a_2) , then $\mathbf{a}'$ have components $(-a_2, a_1)$. Thus, the area is

$$\frac{1}{2} \mathbf{a}' \cdot \mathbf{b} = \frac{1}{2} (-a_2 b_1 + a_1 b_2) = \frac{1}{2} \det \begin{pmatrix} a_1 & a_2 \\ b_1 & b_2 \end{pmatrix}.$$

□

Remark. However, in this way, the area is not necessarily positive. Thus, we adopt the convention of signed area: the area formed by $\mathbf{a}$ rotating counter-clockwise to $\mathbf{b}$ is positive by convention.

By the determinant definitions from the matrices section, the three-dimensional determinant gives the signed volume of the parallelepiped formed by $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$.

Now, we are ready to define cross product.

Definition 2.3.2. For two given three dimensional vectors $\mathbf{a}$ and $\mathbf{b}$ with components (a_1, a_2, a_3) and (b_1, b_2, b_3) , respectively. We define the cross product between $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a} \times \mathbf{b}$, as the vector

$$\det \begin{pmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{pmatrix}. \quad (2.3.4)$$

Theorem 2.3.3. For two given three dimensional vectors $\mathbf{a}$ and $\mathbf{b}$, $\|\mathbf{a} \times \mathbf{b}\|$ is equal to the (signed) area of the parallelogram formed by $\mathbf{a}$ and $\mathbf{b}$. And the direction of $\mathbf{a} \times \mathbf{b}$ is orthogonal to the plane formed by $\mathbf{a}$ and $\mathbf{b}$, while satisfying the right hand rule.

Proof. This result comes directly from the definition of the cross product. $\square$

Theorem 2.3.4. For three 3D vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$,

$$\begin{aligned} \det(\mathbf{a}, \mathbf{b}, \mathbf{c}) &= \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) \\ &= \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a}) \\ &= \mathbf{c} \cdot (\mathbf{a} \times \mathbf{b}). \end{aligned}$$

Proof. This result comes directly from the expansion of both sides. However, an argument based on the geometric meaning in terms of the area of the parallelepiped could also be established. $\square$

Theorem 2.3.5 (LAGRANGE IDENTITY (THREE DIMENSIONS)).

$$\|\mathbf{a} \times \mathbf{b}\|^2 = \|\mathbf{a}\|^2 \|\mathbf{b}\|^2 - (\mathbf{a} \cdot \mathbf{b})^2$$

Proof. Although direct expansion with the component form does verify this identity, we hereby proceed with a more geometric argument.

Let θ be the angle between $\mathbf{a}$ and $\mathbf{b}$, we have

$$\begin{aligned} \|\mathbf{a} \times \mathbf{b}\|^2 &= \|\mathbf{a}\| \|\mathbf{b}\| \sin \theta \\ &= (\|\mathbf{a}\|^2 \|\mathbf{b}\|^2)^{1/2} \cdot \sin^2 \theta \\ &= (\|\mathbf{a}\|^2 \|\mathbf{b}\|^2)^{1/2} \cdot (1 - \cos^2 \theta) \\ &= \|\mathbf{a}\|^2 \|\mathbf{b}\|^2 - (\mathbf{a} \cdot \mathbf{b})^2. \end{aligned}$$

$\square$

Remark. In fact, this theorem is generalizable to finitely many vectors.

$$\left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right) - \left(\sum_{i=1}^n a_i b_i \right)^2 = \sum_{1 \leq i < j \leq n} (a_i b_j - a_j b_i)^2$$

Example 2.3.6 ((18.02 Pset2 PartII P1). Suppose we know that when the three planes P_1 , P_2 , and P_3 in $\mathbb{R}^3$ intersect in pairs, we get three lines L_1 , L_2 , and L_3 which are *distinct* and *parallel*.

- a) Sketch a picture of this situation.
- b) Show that the three normals to P_1 , P_2 , and P_3 all lie in one plane, using a geometric argument.
- c) Show that the three normals to P_1 , P_2 , and P_3 all lie in one plane, using an algebraic argument. (Note that the three planes clearly do *not* all intersect at one point.)

Solution. (a) Graph omitted, the intersection of the three planes form a triangular prism. (c) Let the normals of the three planes be $\mathbf{n}_1$, $\mathbf{n}_2$, and $\mathbf{n}_3$, respectively. The pairwise intersections of the three planes have direction given by $\mathbf{n}_1 \times \mathbf{n}_2$, $\mathbf{n}_2 \times \mathbf{n}_3$, and $\mathbf{n}_3 \times \mathbf{n}_1$. The three lines are parallel gives all of these quantities are equal.

$$\mathbf{n}_1 \times \mathbf{n}_2 = \mathbf{n}_2 \times \mathbf{n}_3 \implies -\mathbf{n}_2 \times \mathbf{n}_1 = \mathbf{n}_2 \times \mathbf{n}_3 \implies -\mathbf{n}_1 \sim \mathbf{n}_3 \implies \mathbf{n}_1 \sim \mathbf{n}_3.$$

Similarly, $\mathbf{n}_2 \sim \mathbf{n}_1$, so all three normals are coplanar. $\square$

Example 2.3.7 (Pset2 PartII P3). For any plane P which is not parallel to the x - y plane, define the *steepest direction* on P to be the direction of any vector which lies in P and which makes the largest (acute) angle with the x - y plane.

a) Let P be the plane through the origin with normal vector $\mathbf{n}$. Derive a formula, in terms of $\mathbf{n}$, for a vector $\mathbf{w}$ which points in the steepest direction on P .

b) Now let P be the plane through the origin which contains two non-parallel vectors $\mathbf{u}$ and $\mathbf{v}$, where $\mathbf{u}$ and $\mathbf{v}$ do not both lie in the x - y plane. Derive a formula, in terms of $\mathbf{u}$ and $\mathbf{v}$, for a vector $\mathbf{w}$ which points in the steepest direction on P .

Example 2.3.8. Fix three 3D dimensional vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$. Determine whether another vector $\mathbf{v}$ is on the plane formed by $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$.

Solution. The vector $\mathbf{v}$ is on the plane iff

$$\mathbf{v} \cdot ((\mathbf{a} - \mathbf{b}) \times (\mathbf{a} - \mathbf{c})) = 0.$$

Symmetrically, it could be expressed as

$$\det(\mathbf{p} - \mathbf{a} \quad \mathbf{p} - \mathbf{b} \quad \mathbf{p} - \mathbf{c}) = 0.$$

$\square$

3 Matrices

There are many motivations for matrices, including transforming coordinate systems and solving systems of linear equations. Here, we begin by looking at it through the lens of linear combinations.

3.1 Multiplication

Definition 3.1.1. For vectors $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$, etc., rewrite its linear combination of

$$(\mathbf{u} \ \mathbf{v} \ \mathbf{w} \ \cdots) \cdot \begin{pmatrix} a \\ b \\ c \\ \vdots \end{pmatrix} = a\mathbf{u} + b\mathbf{v} + c\mathbf{w} + \cdots,$$

where

$$(\mathbf{u} \ \mathbf{v} \ \mathbf{w} \ \cdots)$$

could be written as

$$\begin{pmatrix} u_1 & v_1 & w_1 & \cdots \\ u_2 & v_2 & w_2 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

In this way, a matrix a row vector of vectors.

In this definition,

$$\begin{cases} 2x_1 + 3x_2 + 3x_3 = u_1 \\ 2x_1 + 4x_2 + 5x_3 = u_2 \\ x_1 + x_2 + 2x_3 = u_3 \end{cases}$$

could be written as

$$\begin{pmatrix} 2 & 3 & 3 \\ 2 & 4 & 5 \\ 1 & 1 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}$$

Definition 3.1.2. Let $\mathbf{A} = (a_{ij})$ be an $m \times n$ matrix and $\mathbf{B} = (b_{ij})$ be an $n \times p$ matrix.

The **matrix product** $\mathbf{AB}$ is the $m \times p$ matrix whose (i, j) -entry is

$$(\mathbf{AB})_{ij} = \sum_{k=1}^n a_{ik} b_{kj}.$$

Equivalently, the (i, j) -entry of $\mathbf{AB}$ is obtained by taking the dot product of the i th row of $\mathbf{A}$ with the j th column of $\mathbf{B}$.

The product $\mathbf{AB}$ is defined only when the number of columns of $\mathbf{A}$ equals the number of rows of $\mathbf{B}$.

Aside from the definition, there are several other helpful ways to interpret matrix multiplication.

3.2 Row Picture and Column Picture

When visualizing linear systems, there are often two main ways, corresponding to two ways of interpreting the product between a matrix and a vector.

Example 3.2.1. Solve the system of linear equations

$$\begin{cases} 2x - y = 0 \\ -x + 2y = 3 \end{cases}.$$

Solution. This equation could be written in the form of

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 3 \end{pmatrix}.$$

Let

$$\mathbf{A} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 0 \\ 3 \end{pmatrix},$$

then $\mathbf{Ax} = \mathbf{b}$. This represents two lines on $\mathbb{R}^2$ intersecting at a point, termed by Prof. Strang the “row picture,” where each row of $\mathbf{A}$ combines $\mathbf{x}$, and setting that combination to an element of $\mathbf{b}$ gives an equation, in this case, one that describes a line in $\mathbb{R}^2$.

However, just like “counting twice” (counting the same thing in two different ways) in combinatorics, looking the other way, as the “column picture” produces a different kind of insights. By treating multiplying a vector on the right as a combination of the column vectors of the matrix, the same linear system could also be expressed as

$$x \begin{pmatrix} 2 \\ -1 \end{pmatrix} + y \begin{pmatrix} -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 3 \end{pmatrix}.$$

This equation offers a completely different geometric interpretation: given two vectors, $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ 2 \end{pmatrix}$, for what values of x and y would their linear combination be $\begin{pmatrix} 0 \\ 3 \end{pmatrix}$.

With this “column picture,” it’s natural to ask, given $\mathbf{A}$ what combinations, or $\mathbf{b}$ could be achieved, by varying $\mathbf{x}$. In fact, for reasons that as we shall see

later in vector spaces, any combination could be achieved. In other words, the linear combination of $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ 2 \end{pmatrix}$ fill the entire plane. $\square$

Remark. Notice that the abstraction from the linear system to matrices to a seemingly unnecessary step explicitly representing matrices with letters demonstrates the role that abstraction plays as a useful technique in math. Just like the use of ∇ and representing curl and divergence in terms of which, these key symbols serve as a holder when connecting distinct concepts, one of the main themes of math.

Adding one variable to a system of 3 equations of 3 variables, the “row picture” becomes relatively less easy to visualize and represent. With more variables and equations, this trend continues, visualization already becomes difficult. However, with the equally important, but often missed “column picture,” things are not much harder.

Another contrast between the “row picture” and “column picture” is when defining or computing the product between a vector and a matrix.

Example 3.2.2. Compute the product of the following matrix and vector

$$\begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

Solution. Usually, this is done through the dot product of every row in the matrix and the vector; however, the “column” way takes each column in the matrix as a vector and the product is the summation of the i -th column and the i -th component of the vector.

$$\begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = 1 \begin{pmatrix} 2 \\ 1 \end{pmatrix} + 2 \begin{pmatrix} 5 \\ 3 \end{pmatrix} = \begin{pmatrix} 12 \\ 7 \end{pmatrix}.$$

$\square$

Remark. Thus, the product of a matrix and a vector, Ax is a linear combination of the columns of A .

More generally, fix matrices A , B , and C ,

$$C = AB,$$

then each column of C is a linear combination of the columns of A , with coefficients taken from the corresponding column of B . Likewise, each row of C is a linear combination of the rows of B , with coefficients taken from the corresponding row of A .

3.2.1 out product

Matrix multiplication can also be viewed from a third perspective. Writing the columns of A as $a_1, \dots, a_n$ and the rows of B as $b_1^T, \dots, b_n^T$, we have

$$AB = a_1 b_1^T + a_2 b_2^T + \dots + a_n b_n^T.$$

Thus, the product is a sum of outer products of columns of A and rows of B .

3.2.2 block

Finally, these viewpoints extend naturally to block matrices. If a matrix is partitioned into blocks of compatible sizes, each block may be treated as a single entry, and the usual rules of matrix multiplication remain valid. This allows complicated matrix products to be analyzed and computed using larger building blocks rather than individual entries.

Matrix algebra satisfies many properties that makes it a nice algebraic structure.

1. Associative

Theorem 3.2.2.1. *In other words, if X is a vector and A and B are matrices,*

$$(AB)X = A(BX).$$

This theorem could be interpreted with the lens of matrices as transformations. The transformation AB is equivalent to applying the transformation described by A first, then transformation described by matrix B .

2. Commutative

However, note that $AB \neq BA$, which makes sense after the above theorem. In fact, given AB , BA is not even necessarily well-defined.

3. Multiplicative Identity

There exist an unique identity fixing the height of the matrix being multiplied, with 1 along the diagonal and 0 elsewhere. Denote it as I , we have

$$IX = X.$$

Multiplicative Inverse

Definition 3.2.2.2. For a matrix A , matrix M is its inverse iff $AM = MA = I$, where I is the identity matrix.

Observe that a matrix has an inverse only if it is a square matrix, meaning it has the same number of rows and columns. In fact, a square matrix's left inverse is also its right inverse, and this is a determining property of square matrices.

If a matrix is invertible when it has an inverse, then a matrix $\mathbf{A}$ is not invertible iff there exists a non-zero vector $\mathbf{x}$ that $\mathbf{A}\mathbf{x} = \mathbf{0}$.

Moreover,

$$(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1},$$

as

$$\mathbf{ABB}^{-1}\mathbf{A}^{-1} = \mathbf{AIA}^{-1} = \mathbf{AA}^{-1} = \mathbf{I}.$$

Example 3.2.2.3. Consider the matrix

$$\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Then

$$\mathbf{R} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -y \\ x \end{pmatrix},$$

so $\mathbf{R}$ rotates every vector in the plane by 90° counterclockwise about the origin.

In particular,

$$\mathbf{R} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

and

$$\mathbf{R} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}.$$

Thus, a system described by $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$ is mapped to a system of $\hat{\mathbf{j}}$ and $-\hat{\mathbf{j}}$. Therefore, 90° counterclockwise.

3.3 Transpose

Definition 3.3.1. The transpose of a matrix, $\mathbf{A}$, is $\mathbf{A}^\top$, such that $(\mathbf{A}^\top)_{ij} = \mathbf{A}_{ji}$.

Then, we have the following rules for transpose

1. Addition

$$(\mathbf{A} + \mathbf{B})^\top = \mathbf{A}^\top + \mathbf{B}^\top.$$

2. multiplication

$$\mathbf{AB}^\top = \mathbf{B}^\top \mathbf{A}^\top.$$

The reverse of A and B is due to AB_{ij} is A 's i -th row dot B 's j -th column, which is B 's j -th row dot A 's i -th column.

3. Inverse

$$(\mathbf{A}^{-1})^\top = (\mathbf{A}^\top)^{-1}.$$

It suffices to show

$$(\mathbf{A}^{-1})^\top \mathbf{A}^\top = \mathbf{I}.$$

Therefore, the dot product, also as known as the inner product of two vectors, $\mathbf{x}$ and $\mathbf{y}$ is $\mathbf{x} \cdot \mathbf{y} = \mathbf{x}^\top \mathbf{y}$.

$$\begin{aligned} \mathbf{A}\mathbf{A}^{-1} = \mathbf{I} &\implies (\mathbf{A}\mathbf{A}^{-1})^\top = \mathbf{I}^\top = \mathbf{I} \\ &\implies (\mathbf{A}^{-1})^\top \mathbf{A}^\top = \mathbf{I}. \end{aligned}$$

Since vectors could be viewed as special cases of matrices, transpose is also a pertinent operation regarding vectors.

Definition 3.3.2. Fix vectors $\mathbf{x}$ and $\mathbf{y}$, with dimensions m and n , respectively. In the case where $m = n$, define the inner product between $\mathbf{x}$ and $\mathbf{y}$ to be

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \mathbf{y},$$

which is exactly the dot product between $\mathbf{x}$ and $\mathbf{y}$. By transposing $\mathbf{y}$ instead, define

$$|\mathbf{x}\rangle\langle\mathbf{y}| = \mathbf{x}\mathbf{y}^\top,$$

producing a $m \times n$ matrix.

Interestingly, we have

$$(\mathbf{A}\mathbf{x})^\top \mathbf{y} = \mathbf{x}^\top \mathbf{A}^\top \mathbf{y} = \mathbf{x}^\top (\mathbf{A}^\top \mathbf{y}).$$

Definition 3.3.3. A matrix $\mathbf{A}$ is *symmetric* iff $\mathbf{A}^\top = \mathbf{A}$.

We have the following properties of symmetric matrices.

1. If matrix $\mathbf{A}$ is invertible and symmetric, then $\mathbf{A}^{-1}$ is symmetric.

$$(\mathbf{A}^{-1})^\top = (\mathbf{A}^\top)^{-1} = \mathbf{A}^{-1}.$$

2. $\mathbf{A}\mathbf{A}^\top$ is symmetric

$$(\mathbf{A}\mathbf{A}^\top)^\top = \mathbf{A}\mathbf{A}^\top.$$

3. In elimination, $\mathbf{A} = \mathbf{L}\mathbf{U}$, introduce the diagonal matrix $\mathbf{D}$ such that $\mathbf{L}$ and $\mathbf{U}$ both have all 1's on their diagonals, $\mathbf{A} = \mathbf{L}\mathbf{D}\mathbf{U} = \mathbf{L}\mathbf{D}\mathbf{L}^\top$. In this way,

storage and amount of computation for solving a symmetric linear system are both cut in half.

3.4 Permutation Matrices

Definition 3.4.1. A matrix P is a permutation matrix iff there is exactly one 1 in each of its column and row, and 0 elsewhere.

Following are some observations of permutation matrices

1. all permutation matrices are square matrices
2. $P^\top$ is also a permutation matrix.
3. A permutation matrix times a permutation matrix is still a permutation matrix.
4. There are $n!$ permutation matrices of $n \times n$.
5. $P^{-1} = P^\top$

This is true as, upon transpose, the 1 in each row of P is multiplied with the 1 in the corresponding column of $P^\top$.

note to self: left out at page 122, and still need to reformat the inverse section

Example 3.4.2. (a) If you take powers of a permutation matrix, why is some P^k eventually equal to I ? (b) Find a 5 by 5 permutation matrix P so that the smallest power to equal I is P^6 .

Solution. (a) Since there are only $n!$ permutation matrices of size $n \times n$, there will eventually be a power that repeats a previously appeared matrix, creating a cycle. Meaning

$$P^a = P^b \implies P^{-b} P^a = P^{a-b} = I.$$

(b) A simple “shift right one” matrix has a order of 5 rather than 6. But thinking in terms of cycles directly gives that a cycle of 2 and a cycle of 3 would do the work. So,

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix},$$

would have the highest possible order of 6. □

3.5 Difference and Sum Matrices

Definition 3.5.1. A difference matrix D is a $n \times n$ square matrix where

$$D_{ij} = \begin{cases} 1 & \text{if } i = j \\ -1 & \text{if } i = j + 1. \\ 0 & \text{otherwise} \end{cases}$$

When acted upon a vector $\mathbf{x}$, ie. $\mathbf{Ax}$, each element in the resulting vector represents the difference between the corresponding element in $\mathbf{x}$ and the previous element, with the exception of the first element, which remains the same. Formally,

$$(\mathbf{Ax})_i = \mathbf{x}_i - \mathbf{x}_{i-1}$$

, where $(\mathbf{Ax})_0 := 0$.

Conversely, define the sum matrix as the following.

Definition 3.5.2. A sum matrix $\mathbf{S}$ is a $n \times n$ square matrix where

$$D_{ij} = \begin{cases} 1 & \text{if } j \leq i \\ 0 & \text{if } j > i \end{cases}$$

Then, for a vector $\mathbf{x}$, $\mathbf{Ax}$ is a vector where each element in the resulting vector is the sum of all the elements in $\mathbf{x}$ up to the corresponding element,

$$(\mathbf{Ax})_i = \sum_{k=1}^i \mathbf{x}_k.$$

Theorem 3.5.3. Let $\mathbf{D}_n$ be the difference matrix with n rows and $\mathbf{S}_n$ be the sum matrix with n rows, then $\mathbf{D}_n \mathbf{S}_n = \mathbf{I}$, ie. they are inverses of each other.

By visualizing the vector $\mathbf{x}$ to be dense as a function, say $x(t)$,

$$\mathbf{D}\mathbf{x} = \frac{d}{dt}x(t)$$

and

$$\mathbf{S}\mathbf{x} = \int x(t) dt$$

.

This viewpoint provides another way to look at the relationship between differentiation and integration.

Furthermore, by using rules on matrices, we are able to see properties of differentiation and integration in a linear algebra way.

Definition 3.5.4. Define the inner product of two functions $\mathbb{R} \rightarrow \mathbb{R}$, $f(x)$ and $g(x)$, to be

$$\int_{-\infty}^{\infty} f(x)g(x) \, dx.$$

4 Linear Equations

One of the problems that motivated linear algebra is finding the solution to a system of linear equations, which naturally gives rise to matrices, and several important operations of matrices. In the language of linear combinations, the problem of linear systems is that “given a set of vectors, which linear combinations of them will produce a fixed vector.”

Formally, given coefficient matrix A and vector b , if $Ax = b$, we aim to find x .

To this end, if x could be uniquely determined for all b , term A as *invertible*, otherwise, let A be *singular*.

4.1 Row Picture and Column Picture

When visualizing linear systems, there are often two main ways, corresponding to two ways of interpreting the product between a matrix and a vector.

Example 4.1.1. Solve the system of linear equations

$$\begin{cases} 2x - y = 0 \\ -x + 2y = 3 \end{cases}.$$

Solution. This equation could be written in the form of

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 3 \end{pmatrix}.$$

Let

$$A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad x = \begin{pmatrix} x \\ y \end{pmatrix}, \quad b = \begin{pmatrix} 0 \\ 3 \end{pmatrix},$$

then $Ax = b$. This represents two lines on $\mathbb{R}^2$ intersecting at a point, termed by Prof. Strang the “row picture,” where each row of A combines x , and setting that combination to an element of b gives an equation, in this case, one that describes a line in $\mathbb{R}^2$.

However, just like “counting twice” (counting the same thing in two different ways) in combinatorics, looking the other way, as the “column picture” produces a different kind of insights. By treating multiplying a vector on the right as a combination of the column vectors of the matrix, the same linear system could also be expressed as

$$x \begin{pmatrix} 2 \\ -1 \end{pmatrix} + y \begin{pmatrix} -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 3 \end{pmatrix}.$$

This equation offers a completely different geometric interpretation: given two vectors, $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ 2 \end{pmatrix}$, for what values of x and y would their linear combination be $\begin{pmatrix} 0 \\ 3 \end{pmatrix}$.

With this “column picture,” it’s natural to ask, given $\mathbf{A}$ what combinations, or $\mathbf{b}$ could be achieved, by varying $\mathbf{x}$. In fact, for reasons that as we shall see later in vector spaces, any combination could be achieved. In other words, the linear combination of $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ 2 \end{pmatrix}$ fill the entire plane. $\square$

Remark. Notice that the abstraction from the linear system to matrices to a seemingly unnecessary step explicitly representing matrices with letters demonstrates the role that abstraction plays as a useful technique in math. Just like the use of ∇ and representing curl and divergence in terms of which, these key symbols serve as a holder when connecting distinct concepts, one of the main themes of math.

Adding one variable to a system of 3 equations of 3 variables, the “row picture” becomes relatively less easy to visualize and represent. With more variables and equations, this trend continues, visualization already becomes difficult. However, with the equally important, but often missed “column picture,” things are not much harder.

Another contrast between the “row picture” and “column picture” is when defining or computing the product between a vector and a matrix.

Example 4.1.2. Compute the product of the following matrix and vector

$$\begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

Solution. Usually, this is done through the dot product of every row in the matrix and the vector; however, the “column” way takes each column in the matrix as a vector and the product is the summation of the i -th column and the i -th component of the vector.

$$\begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = 1 \begin{pmatrix} 2 \\ 1 \end{pmatrix} + 2 \begin{pmatrix} 5 \\ 3 \end{pmatrix} = \begin{pmatrix} 12 \\ 7 \end{pmatrix}.$$

$\square$

Remark. Thus, the product of a matrix and a vector, $\mathbf{Ax}$ is a linear combination of the columns of $\mathbf{A}$.

More generally, fix matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$,

$$\mathbf{C} = \mathbf{AB},$$

then each column of C is a linear combination of the columns of A , with coefficients taken from the corresponding column of B . Likewise, each row of C is a linear combination of the rows of B , with coefficients taken from the corresponding row of A .

Matrix multiplication can also be viewed from a third perspective. Writing the columns of A as $a_1, \dots, a_n$ and the rows of B as $b_1^T, \dots, b_n^T$, we have

$$AB = a_1 b_1^T + a_2 b_2^T + \dots + a_n b_n^T.$$

Thus, the product is a sum of outer products of columns of A and rows of B .

Finally, these viewpoints extend naturally to block matrices. If a matrix is partitioned into blocks of compatible sizes, each block may be treated as a single entry, and the usual rules of matrix multiplication remain valid. This allows complicated matrix products to be analyzed and computed using larger building blocks rather than individual entries.

4.2 Elimination

Elimination is an efficient and natural way of solving systems of linear equations.

Example 4.2.1.

Solve the system

$$\begin{cases} x + 2y + z = 2 \\ 3x + 8y + z = 12 \\ 4y + z = 2 \end{cases}$$

Solution. In linear algebra, ideas are expressed in matrices. Thus, we rewrite the system as

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{pmatrix} \quad b = \begin{pmatrix} 2 \\ 12 \\ 2 \end{pmatrix}.$$

The corresponding augmented matrix is obtained by adjoining the solutions column to the coefficient matrix, since that column must be manipulated together with the coefficient columns.

Recursively, we eliminate variables by fixing pivots. For convenience, let (m, n) denote the entry in the m -th row and n -th column.

Let the first pivot be

$$p_1 = (1, 1) = 1.$$

Apply

$$R_2 \rightarrow R_2 - 3R_1.$$

This gives

$$\begin{pmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{pmatrix} \quad \begin{pmatrix} 2 \\ 6 \\ 2 \end{pmatrix}.$$

The second pivot is

$$p_2 = (2, 2) = 2.$$

Apply

$$R_3 \rightarrow R_3 - 2R_2.$$

Then

$$\begin{pmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 5 \end{pmatrix} \quad \begin{pmatrix} 2 \\ 6 \\ -10 \end{pmatrix}.$$

The resulting upper-triangular matrix is

$$U = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 5 \end{pmatrix}.$$

The pivots are 1, 2, 5.

With U , the triangular system is

$$\begin{cases} x + 2y + z = 2 \\ 2y - 2z = 6 \\ 5z = -10 \end{cases}.$$

From the last equation,

$$z = -2.$$

Substituting into the second equation,

$$2y - 2(-2) = 6$$

gives

$$2y + 4 = 6,$$

so

$$y = 1.$$

Finally,

$$x + 2(1) + (-2) = 2,$$

hence

$$x = 2.$$

Therefore

$$(x, y, z) = (2, 1, -2).$$

Equivalently,

$$x = \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}.$$

□

Remark. From this example, it is easy to see when the method works and when it fails. When we reach a pivot that is 0, we may proceed only if there is a lower row with a non-zero entry in that column, in which case we exchange the two rows. If there is no such row, the method fails, and the matrix is non-invertible. **note to self: add the reference to the inverse of a matrix when i finish that section**

However, in linear algebra, we also want to represent the row operations themselves in matrix form. Since we are manipulating rows, the natural object is an elementary matrix multiplying the main matrix on the left.

For example,

$$E_{21} = \begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

encodes the operation $R_2 \rightarrow R_2 - 3R_1$, and

$$E_{32} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{pmatrix}$$

encodes the operation $R_3 \rightarrow R_3 - 2R_2$.

Therefore, the elimination steps may be written as

$$E_{32}E_{21}A = U.$$

Because matrix multiplication is associative, we may group the operations into a single matrix

$$E = E_{32}E_{21},$$

so that

$$EA = U.$$

((type: pagebreak),)

Example 4.2.2.

Let

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

and let

$$r = (1 \ 2 \ 7).$$

Then

$$rA = (1 \ 2 \ 7) \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}.$$

Solution. According to the row-combination rule,

$$rA = 1 \cdot (1 \ 2 \ 3) + 2 \cdot (4 \ 5 \ 6) + 7 \cdot (7 \ 8 \ 9).$$

Computing the linear combination,

$$rA = (1 + 8 + 49 \ 2 + 10 + 56 \ 3 + 12 + 63) = (58 \ 68 \ 78).$$

Direct multiplication gives the same result:

$$(1 \ 2 \ 7) \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} = (58 \ 68 \ 78).$$

□

Remark. Similar to the column picture, the row picture for multiplication is when a row matrix is multiplied with a matrix, producing a linear combination of the rows of that matrix.

From the two examples, we see that elementary matrices act differently depending on which side they multiply a matrix. Left multiplication performs row operations, while right multiplication performs column operations. If E is an elementary matrix, then EA applies a row operation to A , whereas AE applies a column operation to A .

In summary, elimination takes the following steps

1. Elimination goes from A to a triangular U by a sequence of matrix steps E_{ij} .
2. The inverse matrices E_{ij}^{-1} in reverse order bring U back to the original A .
3. In matrix language that reverse order is $A = LU = (\text{lower triangle})(\text{upper triangle})$.
4. Elimination succeeds if A is invertible. (It may need row exchanges.).

5 Vector Spaces and Subspaces

The jump from vectors to vector spaces and subspaces is similar to the jump from numbers to vectors. Thinking in terms of vector spaces provide a higher level of abstraction in linear algebra operations.

“We are looking inside the calculations, to find the mathematics.”

5.1 Spaces of Vectors

Definition 5.1.1. A set which linear combination of a kind of vectors is closed. In other words, a set of vectors satisfying certain requirements.

Example 5.1.2. $\mathbb{R}^2$ is closed for vectors of exactly two real components, or consists all vectors consisting of two real components.

Definition 5.1.3. In fact, there could be a vector space in $\mathbb{R}^2$ that is not a $\mathbb{R}^2$ that is still closed under linear combinations. Define such a vector space to be a proper subspace of $\mathbb{R}^2$.

Notice that every subspace must contain 0 due to scalar multiplication by 0. And notice that every line passing through (0, 0) is also a subspace. Or, the more or less trivial subspace only containing (0, 0) is also a subspace.

Definition 5.1.4. For a matrix A , the vector space formed by the linear combination of its columns is called its *column subspace*, denoted as $C(A)$.

Example 5.1.5. For a matrix $A = \begin{pmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{pmatrix}$, its column space is a plane passing through the origin containing vectors (1, 2, 4) and (3, 3, 1).

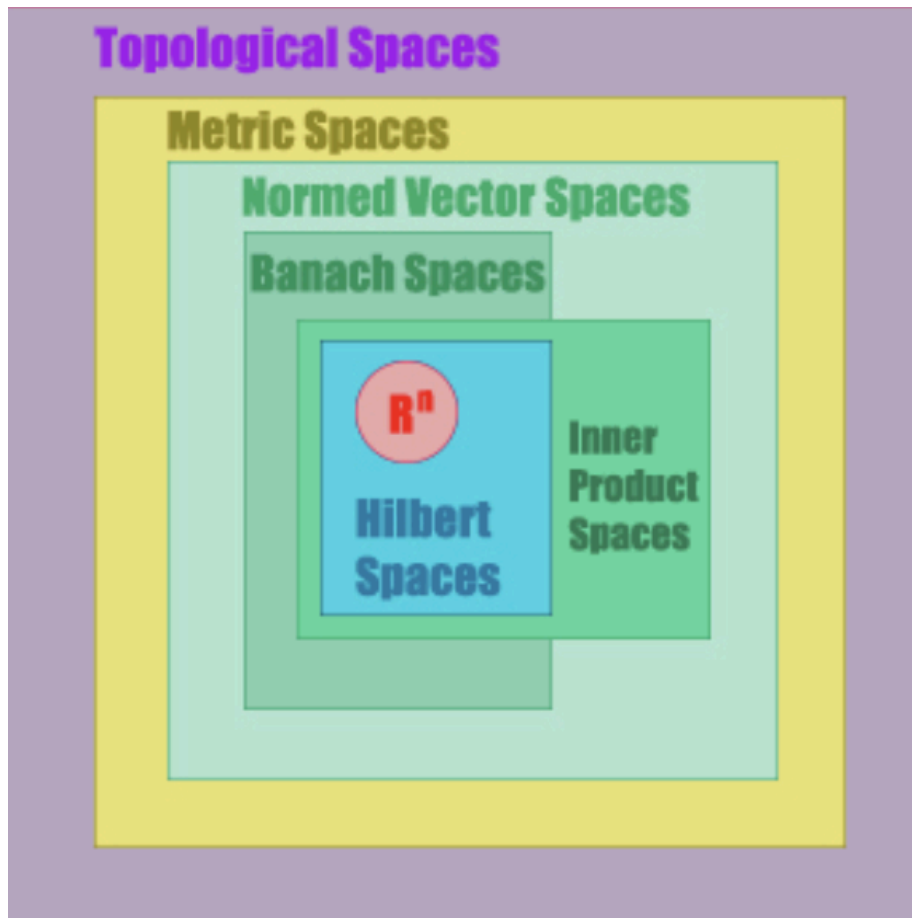
Therefore, notice that $Ax = b$ is solvable iff b is in $C(A)$.

The union of two subspaces wrt the same vector space may not be a subspace, as the sum property is not preserved. However, the union of two subspaces must be a subspace.

In fact, the vector space does not have to be made up of vectors, but any objects that add and (scalar) multiply (or analogs that retain key properties of them) in a closed set. For example, Z is the set containing 0. Also, the set of all real 2×2 real matrices, M is a vector space, and all real functions F is a vector space.

Definition 5.1.6. Generalized, addition must be commutative, associative, has an additive identity, and every element has an inverse; multiplication must be distributive wrt elements and scalar, associative ($c_1(c_2x) = (c_1c_2)x$), and has multiplicative identity.

Here, I must show this photo.



Metric space is defined in Appendix C of my Multivariable Calculus notes.

Definition 5.1.7. For a vector space V , for any set of vectors $S \subseteq V$, the subspace of V spanned by S is defined as the set of linear combinations of S , denoted as SS .

Problem 5.1.1. Suppose S and T are two subspaces of a vector space V . Definition: The sum $S + T$ contains all sums $s + t$, where $s \in S$ and $t \in T$.

(a) Show that $S + T$ satisfies the requirements (addition and scalar multiplication) for a vector space.

(b) If S and T are lines in $\mathbb{R}^m$, what is the difference between $S + T$ and $S \cup T$?

(c) Show that the span of $S \cup T$ is $S + T$.

(d) If $S = C(A)$ and $T = C(B)$, find the matrix M such that $S + T = C(M)$, assume columns of A and B are all in $\mathbb{R}^m$.

Solution. (a) Consider the sum of $s_1 + t_1$ and $s_2 + t_2$, since addition is commutative, this is equal to $(s_1 + s_2) + (t_1 + t_2)$. Since $(s_1 + s_2) \in S$ and $(t_1 + t_2) \in T$, the sum is in $S + T$.

$t_2) \in T$, $(s_1 + s_2) + (t_1 + t_2) \in S + T$. $c(s + t) = cs + ct$ since multiplication is distributive, and due to S and T being subspaces, $cs \in S$ and $ct \in T$, so $c(s + t) \in S + T$.

(b) $S \cup T$ is not necessarily a subspace. For example, S being the x -axis and T being the y -axis.

(c) Any linear combination of an element from S and an element from T could be mapped to an element in $S + T$, in an injection manner. Take $cs + dt$, since $cs \in S$ and $dt \in T$, there is an element in $S + T$ corresponding to it. Additionally, any element in $S + T$ is a linear combination of s and t , specifically $1 \cdot s + 1 \cdot t$.

(d) Notice that S is described by Ax and T described by By , therefore $S + T = Ax + By = M \begin{pmatrix} x \\ y \end{pmatrix}$. Hence, $M = (A \ B)$, also since that the columns of A and B have the same dimensions. $\square$

Remark. In fact, $S + T$ is the smallest subspace containing both S and T .

Problem 5.1.2. Show that the matrices A and $(A \ AB)$ (with extra columns) have the same column space. But find a square matrix with $C(A^2)$ smaller than $C(A)$.

Solution. With the “column picture” of matrix multiplication, AB really is a linear combination of columns of A , which is in $C(A)$.

The matrices I thought of had column space the same as its square; however, gpt hinted on the case where A^2 collapses to 0. For example,

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

whose square is the zero matrix. $\square$

Problem 5.1.3. For what A does an $n \times 11$ matrix has $C(A) = \mathbb{R}^n$.

Solution. $n \leq 11$ and A 's columns are linearly independent. $\square$

5.2 Null Spaces

Definition 5.2.1. For a matrix A , its null space is the set of vectors x such that $Ax = 0$.

Example 5.2.2. Fix matrix

$$A = \begin{pmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{pmatrix},$$

find its null space.

Solution. Notice the solution is any multiples of

$$\begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}.$$

Thus, the null space is a line in $\mathbb{R}^3$. □

Remark. Notice that the zero vector is always in the null space.

Theorem 5.2.3. *The null space is a subspace.*

Proof. We need to show that if $A\mathbf{v} = 0$ and $A\mathbf{w} = 0$, then $A(c\mathbf{v} + d\mathbf{w}) = 0$. Since $A(c\mathbf{v}) = cA\mathbf{v} = 0$, it suffices to show that $A(\mathbf{v} + \mathbf{w}) = 0$.

$$A(\mathbf{v} + \mathbf{w}) = A\mathbf{v} + A\mathbf{w} = 0. \quad \square$$

Looked through the lens of linear systems, when there are more equations than there are variable, it is not generally true that for any $\mathbf{b}$, there are unique solutions. Thus, the null space describes the valid solutions.

5.3 Independence, Basis, Dimensions

Recall the definition of independence from preliminaries, see Definition 2.1.1.

Notice that if the zero vector is in the set of vectors, then they must not be independent.

Consider the matrix A , with columns the set of vectors $\{\mathbf{v}_i\}$, when the number of independent vectors is greater than the dimensions of the vectors, we have free variables, giving a non-zero null space for A . Therefore, we have the following system on dependency.

Theorem 5.3.1. *The n columns of a matrix A are independent iff the null space of A only consists of the zero vector ($\text{rank} = n$), while they are dependent otherwise ($\text{rank} < n$).*

Definition 5.3.2. A basis of a vector space is a set of vectors that are independent and span the space.

Example 5.3.3.

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

is as basis of $\mathbb{R}^3$.

Theorem 5.3.4. The columns of an $n \times n$ matrix $\mathbf{A}$ are a basis of $\mathbb{R}^n$ iff $\mathbf{A}$ is invertible.

Proof. $\mathbf{A}$ is invertible means its n column vectors are independent. Also, n independent vectors of n dimensions span $\mathbb{R}^n$. $\square$

Definition 5.3.5. Fix a vector space, define its *dimension* to be the number of vectors there are in its basis.

Since for a space, the number of vectors its basis has is fixed, its dimension is well-defined (as n independent equations solve for n variables). Thus, the rank of a matrix is the number of its pivot columns, which is the dimension of its column space.

While the dimension of the null space is the number of free variables.

5.4 Fundamental Subspaces

In fact, one of the fundamental objects at study of linear algebra, matrices, have four fundamental subspaces, row space, column space, null space, and left null space. We are going to systematically study them in this section.

Similar to the column space, the row space is the vector space of a matrix's row vectors, or the column space of the matrix's transpose.

Definition 5.4.1. The *row space* of matrix $\mathbf{A}$ is $C(\mathbf{A}^\top)$.

Similar to the relationship between row space and column space, we can define the analogue of the null space. Denoting the null space of a matrix $\mathbf{A}$ to be $N(\mathbf{A})$, we have the following definition.

Definition 5.4.2. The left null space of matrix $\mathbf{A}$ is $N(\mathbf{A}^\top)$.

Fix a matrix $\mathbf{A}$ that is $m \times n$, we wish to understand, for each of these subspaces, what space they are in, how to find their basis vectors, and what's their dimensions.

1. **Column Space** It is in the space of $\mathbb{R}^m$, as each column vector is m dimensional, whereas its dimension is its own rank. Its pivot columns (see the linear equation section) form a basis.
2. **Null Space** By definition, it is in the space of $\mathbb{R}^n$, its dimension is the number of free variables, thus $n - r$
3. **Row Space** Since row space is, by definition, then it is in the space of $\mathbb{R}^n$ and that its dimension is $\text{rank}(\mathbf{A}^\top)$. Moreover, the pivot columns of $\mathbf{A}^\top$ form its basis.
4. **Left Null Space** It is in the space $\mathbb{R}^m$, with dimension $m - r$.

5.5 Matrix Spaces

Vector spaces, are not really spaces of vectors. As long as any objects have a “good” addition and multiplication, ie. satisfying these core properties Definition 5.1.6.

Therefore, the atomic objects at study could very well be matrices.

Example 5.5.1. The space of 3×3 real matrices is a vector space. It has 9 dimensions, each basis could be a matrix with the only non-zero element a 1, each 1 at a distinct position.

Example 5.5.2. The union of the subspace of symmetric matrices S and upper triangular matrices U , both having dimension of 6, is also a subspace, of dimension 3. And, the space of symmetric plus upper triangular, $S + U$, is also a vector space (see Problem 5.1.1), with dimension 9.

Remark. Interestingly,

$$\dim U + \dim S = \dim(U + S) + \dim(U \cap S).$$

Example 5.5.3. Describe the solution space to the second-order ordinary differential equation

$$\frac{d^2 y}{dx^2} + y = 0.$$

Solution. The solutions to this ODE form a vector space. In fact, its two basis vectors are $\cos x$ and $\sin x$. With tools of linear algebra, one may remark that the solution space is 2 dimensional, with basis vectors $\cos x$ and $\sin x$. Therefore, the solutions could be described as $c_1 \cos x + c_2 \sin x$. $\square$

Remark. Thus, one insight stemming from this example is that, to solve a linear differential equation, it suffices to find the basis vectors to its solution space. Also, the dimensions of the solution space is equal to the order of the linear differential equations.

Example 5.5.4. Consider the space of $m \times n$ matrices, does the space formed by all rank 1 matrices form a subspace.

Solution. No. The sum two rank 1 matrices may not be of rank 1. $\square$

Example 5.5.5. Consider four vectors, v_1, v_2, v_3 , and v_4 , show that $\begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{pmatrix}$ forms a vector space and find the basis of this vector space.

Solution. Notice that scalar multiplication preserves the fact that four vectors sum to 0 and addition also preserves that.

Notice that this space is of three dimensions. Also notice that this is the null space of $\mathbf{A}\mathbf{x} = 0$, where

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 & 1 \end{pmatrix}.$$

Since $\dim \mathcal{N}(\mathbf{A}) = n - \text{rank}(\mathbf{A}) \implies \text{rank}(\mathbf{A}) = 1.$ □

6 Applications of Linear Algebra

Linear algebra, though sometimes viewed as primarily a tool for solving linear equations, have many applications in fields such as differential equations, circuits, graphs – perhaps partially due to its high level of abstraction and generality of vector spaces.

6.1 Graphs, Networks, Incidence Matrices

Linear algebra, partially due to its high level of abstraction and generality of vector spaces, have many applications. Here, we examine a discrete example of graphs, and their often useful representation with matrices.

Definition 6.1.1. A *graph* is a set of nodes and edges, such that each edge represents a connection between exactly two nodes. A *directed graph* is a graph where each edge represent a connection that is not symmetric wrt two nodes it is connecting.

To represent this relationship, we represent a graph G , with m edges and n nodes, with an incidence matrix A , defined as following.

Definition 6.1.2. A has m rows and n columns, where an element

$$A_{ij} = \begin{cases} 1 & \text{if edge } i \text{ points to node } j \\ -1 & \text{if edge } i \text{ points away from node } j. \\ 0 & \text{if edge } i \text{ does not connect node } j \end{cases}$$

Now that we have a matrix, many tools from linear algebra is enabled. First, we examine the translation of dependency.

Theorem 6.1.3. *A set of edges forms a loop iff their corresponding row vectors in the incidence matrix of the graph are dependent.*

Given the importance of the four subspaces is to a matrix, we explore the implications of them on the graph.

1. Null space
- 2.

A Appendix

no appendixes so far...